

HOL 2: Package Manager Fundamentals in Linux (Ubuntu & Red Hat)

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Learning Objectives

The objective of this Hands-On Lab (HOL) is to help **beginner-level Linux users** understand:

- What a package manager is
- Why a package manager is required in Linux operating systems
- Different types of package managers used in Linux
- How software is installed, updated, and removed in Ubuntu and Red Hat systems

This lab focuses on **concepts first**, followed by **simple commands**, and then a **clear explanation of what happens in the backend**.

Learning Outcomes

After completing this lab, learners will be able to:

- Clearly explain what a package manager is
 - Understand the role of a package manager in Linux
 - Identify different Linux package managers
 - Install, update, and remove software in Ubuntu using APT
 - Install, update, and remove software in Red Hat using YUM/DNF
 - Confidently perform basic software management tasks on Linux servers
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LAB 1: Package Manager in Ubuntu Linux

Step 1: What Is a Package Manager?

A **package manager** is a system tool that is used to:

- Install software
- Update software
- Remove software
- Manage software dependencies automatically

Without a package manager, users would need to download software manually, resolve dependencies themselves, and manage updates individually.

In Ubuntu, the package manager system is called **APT (Advanced Package Tool)**.

Step 2: Role of a Package Manager in Ubuntu

The package manager in Ubuntu:

- Connects to trusted software repositories
- Downloads verified packages
- Resolves dependencies automatically
- Installs software in correct system locations
- Tracks installed software versions

This ensures system **stability, security, and consistency**.

Step 3: Types of Package Managers (High-Level View)

Linux Family	Package Manager	Package Type
Ubuntu / Debian	apt / apt-get	.deb
Red Hat / CentOS	yum / dnf	.rpm
SUSE	zypper	.rpm
Arch Linux	pacman	.pkg

Step 4: Updating Package Information (Ubuntu)

Command

```
sudo apt update
```

Explanation

This command updates the **local package database** by contacting Ubuntu repositories. No software is installed or upgraded at this stage.

Backend:

- Downloads metadata
 - Updates package list under `/var/lib/apt/lists/`
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Step 5: Installing a Package in Ubuntu

Example: Install `tree`

```
sudo apt install tree
```

Explanation

- APT searches for the package
- Resolves dependencies
- Downloads required `.deb` files
- Installs software using `dpkg`

Verification:

```
tree --version
```

Step 6: Removing a Package in Ubuntu

Remove Package (Keep Config)

```
sudo apt remove tree
```

Full Remove (Remove Config Files)

```
sudo apt purge tree
```

Explanation:

- `remove` deletes binaries
 - `purge` deletes binaries and configuration files
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Step 7: Updating Installed Packages in Ubuntu

```
sudo apt upgrade
```

Explanation: Upgrades all installed packages to the latest available versions based on updated package metadata.

LAB 2 (B) : Package Manager in Red Hat Linux

Step 1: Package Manager in Red Hat Linux

Red Hat-based systems use:

- **YUM** (older versions)
- **DNF** (modern and faster replacement for YUM)

Both tools work with **RPM (.rpm)** packages.

Step 2: Role of Package Manager in Red Hat

The Red Hat package manager:

- Connects to configured repositories
 - Manages software dependencies
 - Ensures enterprise-grade stability
 - Tracks package versions and updates
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Step 3: Updating Package Metadata (Red Hat)

Command

```
sudo dnf makecache
```

Explanation

- Downloads repository metadata
- Updates local package cache
- Prepares system for installation or update

Step 4: Installing a Package in Red Hat

Example: Install `httpd`

```
sudo dnf install httpd
```

Explanation:

- Searches repositories
- Resolves dependencies
- Downloads RPM packages
- Installs using RPM backend

Verification:

```
httpd -v
```

Step 5: Removing a Package in Red Hat

```
sudo dnf remove httpd
```

Explanation:

- Removes binaries
- Configuration files may remain depending on package behavior

Step 6: Updating Packages in Red Hat

```
sudo dnf update
```

Explanation: Updates all installed packages to their latest versions from repositories.

Step 7: How Package Managers Work Internally (Simplified)

1. User runs install/update/remove command
2. Package manager contacts repositories

3. Dependency resolution is performed
 4. Packages are downloaded
 5. Low-level installer (dpkg or rpm) installs software
 6. System database is updated
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End of HOL 2

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